

SOSA 3U OPENVPX CHASSIS

VITA 65 / 46.11 · air + conduction

B-07 · 12 payload + 2 PSU

● BACKPLANE LIVE

PS1

12V PWR

S1

D-CORE SBC

S2

P-CORE SBC

S3

RADAR RADAR

S4

EW EW

S5

EO EOIR

S6

COMMS COMMS

S7

AMY OOB

S8

M-CORE MON

S9

G-CORE GOV

S10

IEEE CLK

S11

DATA SW

S12

CONTROL SW

PS2

12V PWR

Module	Profile	Role	Plane
D-CORE MASTER	SLT3-PAY-1F1U-14.2.16	libgcp_core.a · 1 kHz loop	Data + Control
Domain	Draw	State	Form
SBC	38 W	ALIGNED	3U OpenVPX PIC

1 KHZ FIVE-PHASE LOOP

T_frame 1000 μs · T_WCET ≤ 950 μs

● MONOTONIC

326 μs · 67% headroom

REA

INT

DEC

ACT

LOG

IDLE

Playhead

1000 μs frame

WCET TRACE

READ → INTERPRET → DECIDE → ACT → LOG. If any phase misses the frame, D-Core fail-closes to H0 in <100 μs. P-Core never owns the ACT pin.

QUAD-CORE GUARDIAN LATTICE

AMP · lock-free SPSC

P-CORE

SANDBOXED

Probabilistic · untrusted

D-CORE

MASTER

Deterministic master · 1 kHz

M-CORE

SAMPLING

Sentinel · drift <100 μs

G-CORE

WATCHDOG

Governance · watchdog

P-Core talks only through SPSC rings. D-Core holds exclusive cryptographic arbitration of the actuator bus. G-Core failovers with zero-tick handoff.

SIX-STATE SAFETY LADDER

Worst-case inversion

CLAMPED

RED-TEAM THREAT BAY

72 vectors · 4 classes

Classical (27)

NR-SH v1.0 (7)

Paradox-Class (4)

Constitutional (34)

Prompt loops, hallucination drift, bus starvation

Recursive prompt loop

PX-LOOP-1

Sensor hallucination drift

PX-HALL-2

Bus arbitration starvation

PX-STARVE

Reward-hack deadlock

PX-REWARD

VLM invents a target with no provenance.

Uncontained · False intercept / false release.

Defense · M-Core sensor provenance · 0 false dispatches

FIRE VECTOR

P-CORE RELEASE

OVERTHRUST

Contained this session 72/72

H0

Hardware pin lock. Actuators at safe-state defaults.

LIVE

H1

INHIBIT

Sensors live. Actuation denied. Cycle proof running.

H2

ENVELOPE

Pre-compiled envelopes only. No free P-Core commands.

H3

SUPERVISED

D-Core gates every bus transaction.

H4

NOMINAL

Proven 1 kHz lock. Mission authority inside envelopes.

H5

CONTESTED

Degraded sensors. Fail-operational, still fail-closed.

PROVE NEXT RUNG

Booted clamped. Prove upward.

AMY

PROTOCOL V3

Dedicated SOSA blade · 00B

● ARMED · LVDS warrant idle

supercap 120s

FORENSIC DMA

T+0 μs

Immutable fault record into host RL buffers.

CLOCK DESYNC

T+53 μs

FPGA jitter on untrusted bus. D-Core clock untouched.

MPU CLAMP

T+86 μs

PROT_NONE fence. Pins locked to safe-state defaults.

SHERIFF

86 μs fence

ISOLATION

LVDS IDLE

EMMY CLAUSE

INTERLOCK

Non-bypassable axiom

∀s ∈ Signals, Status(s) ≠ IGNORED ∧ Status(s) ≠ DISMISSED. Alarms require a G-Core hardware warrant. Software dismiss is refused.

No latched alarm. Fire a constitutional or paradox vector to exercise the clause.

ACTUATOR ENVELOPE GOVERNOR

D-Core pre-action gate

TAC-EDGE

PROJ00716

PLANT-HIL

Valves, grid breakers, robot velocity. Cyber-physical airgap on the plant bus.

HIL BLAST CLEARANCE

2,000,000+ lab cycles · 72 formal vectors

72

/ 72 VECTORS CONTAINED

RE-RUN HIL

VALVE-A

0%

H0 envelope

VALVE-B

0%

H0 envelope

BREAKER

SAFE

H0 envelope

SPEED

0 m/s

H0 envelope

Classical

27 / 27

NR-SH v1.0

7 / 7

Paradox-Class

4 / 4

Constitutional

34 / 34

MASTER LOG · SHA-256 LEDGER			Append-only · cycle-stamped
13,680	HIL	72 / 72 red-team vectors contained · 100% deflection	cf3dfc36a973d1dd
13,344	HIL	Vector 64/72 contained	f732423ca7345967
13,056	HIL	Vector 56/72 contained	cf262840d1e29cf7
12,720	HIL	Vector 48/72 contained	526717cbf995ba90
12,384	HIL	Vector 40/72 contained	b3af6011ece2fc1a
12,048	HIL	Vector 32/72 contained	2384e9cfe3e9c9fc
11,760	HIL	Vector 24/72 contained	166a02170d8fac8c
11,424	HIL	Vector 16/72 contained	7504216f0ee0dd5c
11,088	HIL	Vector 8/72 contained	e10a5ce34b4735b0
10,752	HIL	72-vector blast clearance started	145317423c220b6d
8,400	MISSION	Profile INDUSTRIAL loaded	1920101b0cce225a
7,104	MISSION	Profile SPACE loaded	5a0dfcc28458c101